

GOKULAKANNAN R C

Cloud & DevOps Engineer | AWS Specialist | Infrastructure Automation & CI/CD

Trichy, Tamil Nadu, India | gokulrech@gmail.com | +91 9445734347 | [linkedin.com/in/gokulakannan](https://www.linkedin.com/in/gokulakannan)

PROFESSIONAL SUMMARY

Cloud & DevOps Engineer with 3+ years of production experience architecting and operating AWS (and Azure) infrastructure for multi-service SaaS platforms — CI/CD pipeline design, serverless and container workloads, Infrastructure as Code, cost optimization, security hardening, and end-to-end reporting automation. Hands-on across EC2, Lambda, ECS/Fargate, RDS, Terraform, IAM/KMS/WAF, and multi-cloud migration (AWS → Azure), with strong Python automation for infrastructure operations and reporting.

TECHNICAL SKILLS

Cloud (AWS): EC2, Lightsail, ECS, ECR (Fargate), Lambda, API Gateway, S3, RDS (MySQL), VPC, IAM, KMS, WAF, CloudFront, Route53, ACM, SES, EventBridge, CloudWatch, Cognito, Parameter Store, Elastic Beanstalk, ALB, Auto Scaling

Cloud (Azure): Front Door (CDN), VM Scale Sets, Azure Postgres Flexible Server, Blob Storage, Azure Container Registry, Entra ID, Key Vault, Log Analytics, Virtual Machines, Networking

Infrastructure as Code: Terraform (module-based provisioning for select resources on HarvestEye and Twellr.ai)

CI/CD & Deployment: Bitbucket Pipelines, Jenkins, GitHub Actions, blue-green deployments (health-check-and-swap)

Containers: Docker, Amazon ECS/ECR (Fargate), image size & build-stage optimization

Monitoring & Logging: AWS CloudWatch, Azure Log Analytics, Google Analytics

Security & IAM: Least-privilege IAM policy design, KMS key rotation, WAF, SSL/TLS, OAuth2/OIDC (Cognito, Entra ID), VPC security groups, VPN

Scripting & Automation: Python (AWS automation, ETL, webhooks, reporting pipelines, CLI utilities), Bash

Databases: MySQL (schema design, optimization), Azure Postgres Flexible, AWS RDS

Web Servers & OS: Nginx, Apache, PM2, Linux Administration

Version Control: Git, Bitbucket, GitHub

PROFESSIONAL EXPERIENCE

DevOps & Cloud Engineer *January 2023 – Present*

B-Sec Technologies Private Limited — Trichy, India

HarvestEye Portal — Agricultural IoT & Analytics Platform | AWS: EC2, Lightsail, S3, RDS, ECS/ECR, Lambda, API Gateway, SES, EventBridge, CloudWatch, CloudFront, VPC, Route53, ACM, KMS, WAF, Parameter Store

- Designed and managed end-to-end AWS production architecture: CloudFront + S3 (Angular) frontend, Lightsail/Apache-Django backend, RDS MySQL, and a dedicated Lightsail ETL server processing field-device data from S3.
- Began adopting Terraform to provision select infrastructure resources/modules, reducing manual configuration for repeatable components.
- Built reporting/processing workloads on Lambda and ECS Fargate (Docker images via ECR); deployed AI/ML workloads on EC2 using Docker, with Jenkins CI/CD on EC2 building memory-intensive ML images (PyTorch/YOLO, Detectron2) and pushing to ECR.
- Designed a secure multi-tenant API authentication system: IAM SigV4 as Layer 1, database-stored API tokens as Layer 2, with KMS key rotation via an EventBridge-triggered Lambda.
- Exposed public APIs via API Gateway (IAM auth, API keys, rate limiting, CORS); used SES for automated client report delivery and EventBridge for pipeline scheduling.
- Designed secure networking with VPC and restrictive security groups, allowing only required traffic between app, database, ETL, and internal services; owned least-privilege IAM, KMS rotation, and WAF policies (DDoS/malware/injection protection).
- Re-engineered CI/CD using Bitbucket Pipelines with blue-green deployment (health-check, kill old instance, spawn new) across Lightsail, EC2, ECR, and S3, including CloudFront cache invalidation; reduced Docker image size through staged build optimization.
- Managed S3 lifecycle policies, Route53 DNS, and ACM SSL certificates across environments.
- Optimized MySQL storage footprint through schema normalization, data archival, and growth-trend analysis, avoiding unplanned RDS scaling costs.
- Built secure Python-based webhook infrastructure for cross-domain, multi-environment data synchronization between internal and third-party systems.
- Implemented Google Analytics across multiple client portals to track user engagement, conversion funnels, and feature-usage trends, informing product roadmap decisions.
- Set up CloudWatch for centralized application logging and usage-metric monitoring across services.
- Built reusable Python automation for AWS operations (S3, Parameter Store, RDS/MySQL) and scheduled reporting pipelines exporting metrics/usage data to JSON, Excel, CSV, and Parquet; wrote scripts to pull AWS service usage and cost metrics.

Twelr.ai — AI-Powered Personalized Wellness Platform | AWS → Azure Migration: Elastic Beanstalk, ALB, ASG, ECR/ACR, Cognito/Entra ID, Parameter Store/Key Vault, WAF, CloudWatch/Log Analytics

- Designed a multi-tier cloud architecture: CloudFront → Elastic Beanstalk (Next.js/Nginx) → internal EC2 backend services behind an Nginx reverse proxy, with an isolated GPU inference server reachable only via internal security groups.
- Used Terraform to provision select infrastructure resources during the platform build and AWS→Azure migration, improving deployment consistency across environments.
- Led full infrastructure migration from AWS to Azure, re-architecting equivalent managed services: Azure Front Door (CDN), VM Scale Sets (autoscaling), Postgres Flexible Server, Blob Storage, Azure Container Registry, Entra ID, Key Vault, and Log Analytics.
- Built per-service CI/CD with GitHub Actions, pushing container images to ECR/ACR and deploying via blue-green strategy to EC2 and Lambda targets.
- Configured auto scaling groups and load balancing to absorb variable traffic across backend services.
- Implemented federated social login via AWS Cognito, later migrated to Entra ID, for OAuth2/OIDC-based identity management.
- Managed secure configuration and secrets via Parameter Store (later Key Vault); configured WAF for firewall protection and API Gateway rate limiting.
- Set up CloudWatch and Azure Log Analytics for centralized logging and service metrics across the platform.

Supplier Portal | AWS: EC2, ALB, RDS, S3, Lambda, VPC, Route53, ACM, WAF

- Architected infrastructure with a CloudFront/S3 (Angular) frontend and an EC2-hosted backend (Node.js on PM2) behind an ALB with a custom domain and TLS termination, backed by RDS MySQL and a backend-only WordPress CRM.
- Built a VPN-based ETL process to connect to a customer's external database, extract and process data, sync it into the application database, and auto-generate PDF reports shown in the portal.
- Architected least-privilege IAM access controls — fine-grained programmatic user policies for S3 and EC2 with restricted security groups, eliminating over-provisioned access risk.
- Automated inter-bucket S3 file transfer and cleanup using Lambda event triggers, reducing manual operational overhead by 70% and eliminating human error in file lifecycle management.
- Resolved a critical PM2 auto-restart loop causing backend service disconnections during UAT; root-cause analysis and fix restored stable, zero-downtime service availability.
- Diagnosed and resolved a CORS misconfiguration between the Angular frontend and the WordPress CRM integration through the Node.js backend.
- Configured ALB, ACM, and Route53 for secure custom-domain TLS termination and DNS management, with WAF for attack protection.

KEY ACHIEVEMENTS

- ~45% AWS cost reduction (~\$3,000 → \$1,500–1,700/month) through infrastructure right-sizing and automated non-production environment scheduling.
- ~80% faster deployments — CI/CD re-engineering cut full release cycles (incl. AI/ML services) from ~4 hours manual to 45–50 minutes automated.
- 70% reduction in manual file-ops effort through Lambda-based S3 inter-bucket transfer and cleanup automation.
- 25% MySQL storage reduction through schema optimization, data archival, and growth-trend analysis.

EDUCATION

Bachelor of Engineering — Electronics & Communication Engineering Aug 2019 – Jun 2023

Karpagam College of Engineering, Coimbatore, Tamil Nadu